

# Graph on the Coordinate Plane

## Lesson 9-1

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

## Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*A grid with a horizontal line (x-axis) crossing a vertical line (y-axis), making four sections*

### Coordinate plane

A grid with a line going across and a line going up that cross.

*(3, 5) means move right 3 from the origin, then up 5*

### Ordered pair

Two numbers (x, y) that tell where a point is on a grid.

*The starting point at the center where both axes cross — (0, 0)*

### Origin

The point (0, 0) where the two grid lines cross.

*I (+,+) top-right, II (-,+) top-left, III (-,-) bottom-left, IV (+,-) bottom-right*

### Quadrant

One of the four parts of a coordinate grid.

*(3, 2) reflected over the x-axis becomes (3, -2) — same x, opposite y*

### Reflection

A flipped, mirror image across a line.

*..., -3, -2, -1, 0, 1, 2, 3, ...*

### Integer

Whole numbers and their opposites, like -2, -1, 0, 1, 2.

## Key Ideas & Notes



### WHAT WE'RE LEARNING TODAY

**I can plot and identify points on the coordinate plane using ordered pairs.**

#### 1 Notice

Captain Vega left a treasure map, but instead of an 'X marks the spot,' she hid the directions as ordered pairs!

#### 2 Key idea

I can plot and identify points on the coordinate plane using ordered pairs.

#### 3 Words I'll use

Coordinate plane

Ordered pair

Origin

Quadrant

Reflection

Integer



#### Think About It

- What shape will the three points make when connected?
- Which number tells you how far right/left? Which tells you up/down?
- Why is the ORDER in an ordered pair important?



#### Watch

Watch your teacher model one example. Jot what you see.

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#### We try

Solve the next one together as a class.

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#### You try

Now try one on your own.

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## Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

### LAUNCH

**On the coordinate plane, why do we always start at the origin (0, 0) and move sideways before we move up or down to plot a point?**

#### Level 1 support

**Start here:** We start at the origin and move along the x-axis first because \_\_\_\_.

 **Need a hint?**

**Sentence starters (optional)**

**The first number in (x, y) tells me to move \_\_\_\_.**

*El primer numero en (x, y) me dice que me mueva \_\_\_\_.*

**The second number tells me to move \_\_\_\_.**

*El segundo numero me dice que me mueva \_\_\_\_.*

**Word bank:** coordinate plane origin ordered pair x-axis y-axis

**Listen for:** Student states the x-coordinate is the sideways (horizontal) move and the y-coordinate is the up/down (vertical) move, both starting from the origin.

#### Level 2

**Push students to explain why (3, 5) and (5, 3) land on different points even though they use the same numbers.**

- (3, 5) and (5, 3) are different because \_\_\_\_.
- The order in an ordered pair matters because \_\_\_\_.

EXPLORE

**You need to plot (4, 2). Walk your partner through every move you make from the origin, using the words x-axis and y-axis.**

Level 1 support

**Start here:** Starting at the origin, I first move \_\_\_\_, then I move \_\_\_\_.



**Need a hint?**

**Sentence starters (optional)**

**I move \_\_\_\_ units along the x-axis.**

*Me muevo \_\_\_\_ unidades por el eje x.*

**Then I move \_\_\_\_ units along the y-axis.**

*Luego me muevo \_\_\_\_ unidades por el eje y.*

**Word bank:** **origin** **ordered pair** **x-axis** **y-axis** **coordinate plane**

**Listen for:** Student moves 4 right along the x-axis and 2 up along the y-axis, naming each axis and the origin as the start.

Level 2

**Ask students to predict which quadrant the point (4, 2) lands in and justify their answer using the signs of the coordinates.**

- (4, 2) lands in quadrant \_\_\_\_ because \_\_\_\_.
- I can tell the quadrant from the coordinates by \_\_\_\_.

CONNECT

**How is reading a coordinate pair like reading directions on a map or in a video game?**

Level 1 support

**Start here:** An ordered pair works like map directions because \_\_\_\_.



**Need a hint?**

**Sentence starters (optional)**

**The x-coordinate is like going \_\_\_\_.**

*La coordenada x es como ir \_\_\_\_.*

**The y-coordinate is like going \_\_\_\_.**

*La coordenada y es como ir \_\_\_\_.*

**Word bank:** **ordered pair** **coordinate plane** **x-axis** **y-axis** **origin**

**Listen for:** Student connects the *x*-value to horizontal travel and the *y*-value to vertical travel, like map or grid directions from a starting point.

Level 2

**Push students to generalize a rule for plotting ANY ordered pair, including ones with zero in them.**

- A rule that always works for plotting  $(x, y)$  is \_\_\_\_.
- If one coordinate is 0, the point lands on \_\_\_\_ because \_\_\_\_.

REFLECT

A classmate plotted (2, 6) but moved up 2 and over 6. What mistake did they make, and how would you coach them?

Level 1 support

**Start here:** The mistake was \_\_\_\_, and to fix it they should \_\_\_\_.



Need a hint?

Sentence starters (optional)

**They mixed up the \_\_\_\_ and the \_\_\_\_.**

*Confundieron la \_\_\_\_ y la \_\_\_\_.*

**The correct way is to move \_\_\_\_ first.**

*La forma correcta es moverse \_\_\_\_ primero.*

**Word bank:** ordered pair x-axis y-axis origin coordinate plane

**Listen for:** Student identifies that the  $x$ - and  $y$ -coordinates were swapped and explains the correct order ( $x$  first, then  $y$ ).

Level 2

**Ask students to critique why this swap error is so common and suggest a strategy to avoid it every time.**

- This error happens a lot because \_\_\_\_.
- A strategy to remember the order is \_\_\_\_.

## Guided Examples

### Example 1

**Which ordered pair names a point 3 units right and 7 units up from the origin?**

**Solution:** The first coordinate is horizontal (right 3), the second is vertical (up 7). So (3, 7).

**Answer:** A. (3, 7)

### Example 2

**What are the coordinates of the origin?**

**Solution:** The origin is where the x-axis and y-axis cross, at (0, 0).

**Answer:** A. (0, 0)

### Example 3

**A point is at (6, 2). What does the 6 tell you?**

**Solution:** The first number (x-coordinate) tells you horizontal movement. 6 means go right 6 units from the origin.

**Answer:** A. Move 6 units to the right

## Write About the Math

### The Writing Revolution

I can explain my work using the words coordinate plane, ordered pair, origin, and quadrant.

#### 1. Kernel Sentence subject + verb

**Model:** Coordinate plane is a grid with a line going across and a line going up that cross.  
*Plano cartesiano es una cuadrícula con una línea horizontal y una vertical que se cruzan.*

**Write a kernel sentence about coordinate plane. Use a subject and a verb.**

*Escribe una oración base sobre plano cartesiano. Usa un sujeto y un verbo.*

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#### 2. Sentence Expansion because • but • so

**Kernel:** Coordinate plane matters in math  
*Plano cartesiano importa en matemáticas*

Expand the kernel three ways. Add a reason, a contrast, and a result.

**because**  
*porque*

**Coordinate plane matters in math because \_\_\_\_.**  
*Plano cartesiano importa en matemáticas porque \_\_\_\_.*

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**but**  
*pero*

**Coordinate plane matters in math, but \_\_\_\_.**  
*Plano cartesiano importa en matemáticas, pero \_\_\_\_.*

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**so**  
*entonces*

**Coordinate plane matters in math, so \_\_\_\_.**  
*Plano cartesiano importa en matemáticas, entonces \_\_\_\_.*

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### 3. Sentence Types 4 ways to write a math idea

**Statement**  
*Afirmación*

Tell one true fact about coordinate plane.  
*Di un hecho verdadero sobre coordinate plane.*

**Coordinate plane** \_\_\_\_.

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**Question**  
*Pregunta*

Ask a question about coordinate plane.  
*Haz una pregunta sobre coordinate plane.*

**How does** \_\_\_\_ ?

*¿Cómo* \_\_\_\_ ?

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**Exclamation**  
*Exclamación*

Show excitement about coordinate plane.  
*Muestra entusiasmo sobre coordinate plane.*

**Wow,** \_\_\_\_ !

*¡Guau,* \_\_\_\_ !

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**Command**  
*Mandato*

Tell a partner what to do with coordinate plane.  
*Dile a un compañero qué hacer con coordinate plane.*

**First,** \_\_\_\_ .

*Primero,* \_\_\_\_ .

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### 4. Explain Your Reasoning use a sentence starter

**The first number in (x, y) tells me to move** \_\_\_\_.

*El primer numero en (x, y) me dice que me mueva* \_\_\_\_.

**The second number tells me to move** \_\_\_\_.

*El segundo numero me dice que me mueva* \_\_\_\_.

**I move** \_\_\_\_ **units along the x-axis.**

*Me muevo* \_\_\_\_ *unidades por el eje x.*

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## **Try It**

Solve on your own. Check the answer key when you are done.

### **1. What are the coordinates of the origin?**

- A. (0, 0)
- B. (1, 1)
- C. (0, 1)
- D. (1, 0)

Show your work:

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### **2. A point is at (6, 2). What does the 6 tell you?**

- A. Move 6 units to the right
- B. Move 6 units up
- C. Move 6 units to the left
- D. The point is in Quadrant 6

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**A student says that  $(3, 5)$  and  $(5, 3)$  are the same point because they use the same numbers. Is the student correct? Explain why or why not, and describe where each point is located on the coordinate plane.**

*Sentence starter: The student is \_\_\_\_ because  $(3, 5)$  means \_\_\_\_ while  $(5, 3)$  means \_\_\_\_ . The ORDER matters because \_\_\_\_ .*

Show your work:

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## Reflect — Exit Ticket

**Point P is located 6 units right and 3 units up from the origin. What ordered pair names point P?**

- A.  $(6, 3)$
- B.  $(3, 6)$
- C.  $(6, 0)$
- D.  $(0, 3)$

Your answer:

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## Answer Key & Teacher Guide

1. **Try It 1:** A.  $(0, 0)$  — *The origin is where the  $x$ -axis and  $y$ -axis cross, at  $(0, 0)$ .*
2. **Try It 2:** A. Move 6 units to the right — *The first number ( $x$ -coordinate) tells you horizontal movement. 6 means go right 6 units from the origin.*
3. **Exit Ticket:** A.  $(6, 3)$  — *Right 6 =  $x$ -coordinate of 6, Up 3 =  $y$ -coordinate of 3. The ordered pair is  $(6, 3)$ .*

### Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Coordinate plane is a grid with a line going across and a line going up that cross.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).